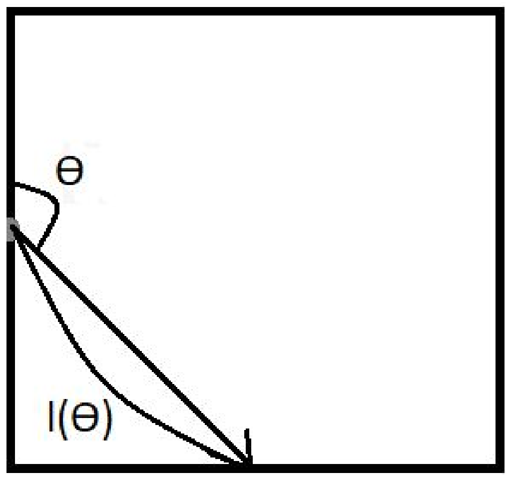
Location-Scale Families

1. CB 3.36
2. CB 3.39

Monte Carlo Integration

3. Imagine a unit square. A ball is rolled in a straight line from the middle of the left side with an angle θ (relative to left side), where $\theta \sim \text{Unif}(0, \pi)$. We are interested in the expected length of the the ball's path before exiting the unit square, denoted $l(\theta)$. The diagram below illustrates one such path.



   (a) Compute the expected value of $l(\theta)$ analytically?

   (b) Confirm that this expected value is correct through Monte Carlo integration using $M = 1000$, 10000, and 100000 samples. Include standard errors for each estimate. Please append code here or at the end of the assignment.

4. Let X be distributed as a truncated Cauchy distribution with location parameter 0, scale parameter 1/5, and truncation between 0 and 1. That is,

$$f(x) = \begin{cases} \frac{5}{\arctan(5)(1+25x^2)} & \text{for } x \in [0,1] \\ 0 & \text{otherwise} \end{cases}$$

   (a) Using $M = 10^4$ samples, estimate $\mathbb{E}(X)$ using Monte Carlo integration with uniform random variables.

   (b) Using $M = 10^4$ samples, estimate $\mathbb{E}(X)$ using Monte Carlo integration by simulating Cauchy random variables.

   (c) Using $M = 10^4$ samples, estimate $\mathbb{E}(X)$ using importance sampling with a Beta(2,1) distribution as an instrumental distribution.

   (d) Using $M = 10^4$ samples, estimate $\mathbb{E}(X)$ using importance sampling with a Beta(1,2) distribution as an instrumental distribution.

   (e) Which estimate [(a) - (d)] has the lowest variance? Why?

Multivariate Probability Calculation

5. CB 4.1
6. CB 4.4abc
7. CB 4.5
8. CB 4.7
9. Let X and Y have the joint density

$$f(x,y) = \frac{6}{7}(x+y)^2, \quad 0 \le x \le 1, \quad 0 \le y \le 1$$

   (a) By integrating over the appropriate regions, find:
      i. $P(X > Y)$.
      ii. $P(X + Y \le 1)$.
      iii. $P(X \le \frac{1}{2})$.

   (b) Find the marginal densities of X and Y.

Conditional Distributions

10. CB 4.31
11. For each of the following, find the marginal distribution of X, $f_X(x)$.
    (a) $X|Y \sim \text{Binomial}(n, Y)$ and $Y \sim \text{Beta}(a, b)$
    (b) $X|Y \sim \text{Exponential}(Y)$ and $Y \sim \text{Gamma}(a, b)$
    (Use rate parametrization for both exponential and gamma)
12. Let X and Y be random variables such that X has density function

$$f_x(x) = \begin{cases} 24x^2 & \text{for } 0 < x < \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$$

    and the conditional density of Y given $X = x$ is

$$f(y|x) = \begin{cases} \frac{y}{2x^2} & \text{for } 0 < y < 2x \\ 0 & \text{otherwise} \end{cases}$$

    What is the conditional density of X given $Y = y$?
13. Let X and Y be continous random variables with joint density function

$$f(x,y) = \begin{cases} 24xy & \text{for } x > 0;\ y > 0;\ 0 < x + y < 1 \\ 0 & \text{otherwise} \end{cases}$$

    What is the conditional probability $\text{P}\left(X < \frac{1}{2} \middle| Y = \frac{1}{4}\right)$?
14. Let $X|Y \sim \text{Pareto}(1, Y)$ and $Y \sim \text{Gamma}(\alpha, \beta)$ where β is the rate and not the scale. Find the marginal pdf of X.

Working With Distributions

15. CB 3.24 (parts a, b, and d; you only need to find the form of the pdf. You do not need to verify it is a pdf or calculate the mean and variance.)

16. Let the independent random variables X_1, X_2, and X_3 have the same pdf:

$$f(x) = 3x^2 \text{ for } 0 < x < 1, \text{ and zero elsewhere.}$$

Find the probability that exactly two of the three variables exceed (1/2).

17. Suppose the joint pdf of the lifetimes of a certain part and a spare is given by

$$f(x, y) = e^{-(x+y)}, \quad 0 < x < \infty, \quad 0 < y < \infty$$

and zero otherwise. Find each of the following:

(a) The marginal pdf's, $f_1(x)$ and $f_2(y)$.

(b) The joint CDF, $F(x, y)$.

(c) $P(X > 2)$.

(d) $P(X < Y)$.

(e) $P(X + Y > 2)$.

(f) Are X and Y independent?